

# Olympiad Inequalities

Vin

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## Disclaimer

These notes are a compact compilation of inequalities encountered during Olympiad preparation. They are intended for quick revision and as a reference to some of the most useful and common inequalities in Olympiad problem solving, not as a complete learning resource. For proofs and further inequalities, see

[https://artofproblemsolving.com/wiki/index.php/Inequality#List\\_of\\_Theorems](https://artofproblemsolving.com/wiki/index.php/Inequality#List_of_Theorems)

## Arithmetic–Geometric Mean Inequality

For  $a_1, a_2, \dots, a_n \geq 0$ ,

$$\frac{a_1 + a_2 + \dots + a_n}{n} \geq \sqrt[n]{a_1 a_2 \dots a_n}.$$

Equality iff

$$a_1 = a_2 = \dots = a_n.$$

## Weighted AM–GM Inequality

For  $a_1, a_2, \dots, a_n \geq 0$  and weights  $\lambda_1, \lambda_2, \dots, \lambda_n > 0$ ,

$$\frac{\lambda_1 a_1 + \lambda_2 a_2 + \dots + \lambda_n a_n}{\lambda_1 + \lambda_2 + \dots + \lambda_n} \geq \left( a_1^{\lambda_1} a_2^{\lambda_2} \dots a_n^{\lambda_n} \right)^{1/(\lambda_1 + \lambda_2 + \dots + \lambda_n)}.$$

Equality iff

$$a_1 = a_2 = \dots = a_n.$$

## Power Mean Inequality

For  $a_1, a_2, \dots, a_n > 0$  and nonzero real numbers  $k_1 > k_2$ ,

$$\left( \frac{1}{n} \sum_{i=1}^n a_i^{k_1} \right)^{1/k_1} \geq \left( \frac{1}{n} \sum_{i=1}^n a_i^{k_2} \right)^{1/k_2}.$$

Equality iff

$$a_1 = a_2 = \dots = a_n.$$

## Cauchy–Schwarz Inequality

For real  $a_i, b_i$ ,

$$\left(\sum_{i=1}^n a_i b_i\right)^2 \leq \left(\sum_{i=1}^n a_i^2\right) \left(\sum_{i=1}^n b_i^2\right).$$

Equality iff the vectors  $(a_1, a_2, \dots, a_n)$  and  $(b_1, b_2, \dots, b_n)$  are linearly dependent.

## Titu's Lemma

For real  $a_i$  and  $b_i > 0$ ,

$$\frac{a_1^2}{b_1} + \frac{a_2^2}{b_2} + \dots + \frac{a_n^2}{b_n} \geq \frac{(a_1 + a_2 + \dots + a_n)^2}{b_1 + b_2 + \dots + b_n}.$$

Equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \dots = \frac{a_n}{b_n}.$$

## Chebyshev's Inequality

If

$$\begin{aligned} a_1 &\leq a_2 \leq \dots \leq a_n, \\ b_1 &\leq b_2 \leq \dots \leq b_n, \end{aligned}$$

then

$$\frac{1}{n} \sum_{i=1}^n a_i b_i \geq \left(\frac{1}{n} \sum_{i=1}^n a_i\right) \left(\frac{1}{n} \sum_{i=1}^n b_i\right).$$

If the orders are opposite, the inequality reverses. Equality iff one sequence is constant.

## Rearrangement Inequality

If

$$\begin{aligned} a_1 &\leq a_2 \leq \dots \leq a_n, \\ b_1 &\leq b_2 \leq \dots \leq b_n, \end{aligned}$$

then for any permutation  $\pi$ ,

$$\sum_{i=1}^n a_i b_{n+1-i} \leq \sum_{i=1}^n a_i b_{\pi(i)} \leq \sum_{i=1}^n a_i b_i.$$

If both sequences are strictly increasing, right equality iff  $\pi(i) = i$ ; left equality iff  $\pi(i) = n + 1 - i$ .

## Hölder's Inequality

For real  $a_i, b_i$  and  $p, q > 1$  with  $\frac{1}{p} + \frac{1}{q} = 1$ ,

$$\sum_{i=1}^n |a_i b_i| \leq \left(\sum_{i=1}^n |a_i|^p\right)^{1/p} \left(\sum_{i=1}^n |b_i|^q\right)^{1/q}.$$

Equality iff one vector is zero or

$$|a_i|^p = \lambda |b_i|^q \quad (i = 1, 2, \dots, n)$$

for some  $\lambda > 0$ .

## Minkowski's Inequality

For real  $a_i, b_i$  and  $p > 1$ ,

$$\left( \sum_{i=1}^n |a_i + b_i|^p \right)^{1/p} \leq \left( \sum_{i=1}^n |a_i|^p \right)^{1/p} + \left( \sum_{i=1}^n |b_i|^p \right)^{1/p}.$$

Equality iff one vector is zero or

$$a_i = \lambda b_i \quad (i = 1, 2, \dots, n)$$

for some  $\lambda \geq 0$ .

## Jensen's Inequality

For convex  $f$ ,  $x_i$  in its domain,  $\lambda_i \geq 0$ , and  $\sum_{i=1}^n \lambda_i = 1$ ,

$$f \left( \sum_{i=1}^n \lambda_i x_i \right) \leq \sum_{i=1}^n \lambda_i f(x_i).$$

For concave  $f$ , the inequality reverses. If  $f$  is strictly convex, equality iff all positive-weight  $x_i$  are equal.

## Bernoulli's Inequality

For  $x > -1$ ,

$$(1+x)^r \geq 1+rx \quad (r \geq 1 \text{ or } r \leq 0),$$
$$(1+x)^r \leq 1+rx \quad (0 \leq r \leq 1).$$

For  $r \neq 0, 1$ , equality iff  $x = 0$ . For  $r = 0$  or  $r = 1$ , equality always holds.

## Schur's Inequality

For  $x, y, z \geq 0$  and  $r > 0$ ,

$$x^r(x-y)(x-z) + y^r(y-z)(y-x) + z^r(z-x)(z-y) \geq 0.$$

Equality iff

$$x = y = z,$$

or, after permutation,

$$(x, y, z) = (t, t, 0) \quad (t \geq 0).$$

For  $r = 1$ ,

$$x^3 + y^3 + z^3 + 3xyz \geq x^2y + x^2z + y^2x + y^2z + z^2x + z^2y.$$

## Muirhead's Inequality

Let

$$\alpha = (\alpha_1, \alpha_2, \dots, \alpha_n), \quad \beta = (\beta_1, \beta_2, \dots, \beta_n)$$

be nonincreasing tuples of nonnegative real numbers. We say that  $\alpha$  majorizes  $\beta$ , written  $\alpha \succ \beta$ , if

$$\sum_{i=1}^k \alpha_i \geq \sum_{i=1}^k \beta_i \quad (1 \leq k < n), \quad \sum_{i=1}^n \alpha_i = \sum_{i=1}^n \beta_i.$$

If  $x_1, x_2, \dots, x_n \geq 0$  and  $\alpha \succ \beta$ , then

$$\sum_{\text{sym}} x_1^{\alpha_1} x_2^{\alpha_2} \cdots x_n^{\alpha_n} \geq \sum_{\text{sym}} x_1^{\beta_1} x_2^{\beta_2} \cdots x_n^{\beta_n}.$$

For positive  $x_i$  and  $\alpha \neq \beta$ , equality iff

$$x_1 = x_2 = \cdots = x_n.$$

### Nesbitt's Inequality

For  $a, b, c > 0$ ,

$$\frac{a}{b+c} + \frac{b}{c+a} + \frac{c}{a+b} \geq \frac{3}{2}.$$

Equality iff

$$a = b = c.$$

### Radon's Inequality

For  $a_i \geq 0$ ,  $b_i > 0$ , and  $m > 0$ ,

$$\sum_{i=1}^n \frac{a_i^{m+1}}{b_i^m} \geq \frac{(a_1 + a_2 + \cdots + a_n)^{m+1}}{(b_1 + b_2 + \cdots + b_n)^m}.$$

Equality iff

$$\frac{a_1}{b_1} = \frac{a_2}{b_2} = \cdots = \frac{a_n}{b_n}.$$